3D Modelling of Space Shuttle

CS302N: Computer Graphics & Visualization Lab

Project Report

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1. Introduction

This project is a 3D Space Shuttle modelling project built using OpenGL, freeglut, and C++. It focuses on creating a interactive model of the NASA Space Shuttle stack, including the Orbiter, External Tank (ET), and Solid Rocket Boosters (SRBs). The project also includes the interior of the Orbiter, such as the flight deck, mid-deck, and payload bay, so the shuttle can be viewed both from the outside and from inside. The model can be controlled using keyboard and mouse input. The project makes use of concepts such as transformations, camera control, lighting, shading, and animation.

The project was developed in five iterations, with each iteration adding a new layer of detail and functionality. In the first iteration, the basic 3D scene was created by setting up the perspective view and orbit camera controls. The Orbiter body was built from simple OpenGL primitives, and basic view options such as wireframe mode, preset viewpoints, and axis lines were added.

In the second iteration, the exterior of the full Space Shuttle stack was completed. The External Tank and both SRBs were added in their correct positions, along with attachment struts, OMS pods, SSME nozzles, the crew cabin area, and the payload bay outline. At this stage, part highlighting was also implemented to help visualize individual components more clearly.

The third iteration focused on lighting and shading. OpenGL lighting was enabled, normals were defined for the shapes, and material properties were assigned to the different shuttle parts. This made the model look more realistic by showing how light falls on the surface of each component. Options to switch between flat and smooth shading, as well as to toggle lighting on and off, were also added for comparison.

In the fourth iteration, the interior of the Orbiter was developed. The flight deck, mid-deck, and payload bay interiors were modelled with simple elements such as seats, panels, lockers, and window frames. A camera mode toggle was added so the user could switch between exterior and interior views. The payload bay doors were also animated, allowing them to open and close during the simulation.

In the fifth iteration, a starfield background with stars, galaxies, asteroids and meteors was added to improve the scene visually, along with a 2D HUD that displays important status information such as camera mode, lighting, shading, and door state. A togglable flame animation was added to the SSME nozzles and SRBs. The external tank was given a texture and a togglable fog option was added to the background. Smooth camera transitions between preset views were implemented. Finally, a display text was added to the fuselage.

2. Specifications

The project was developed using the following technologies:

- **C++17** for the source code
- **OpenGL** for 3D rendering
- **freeglut** for window creation, input handling, and event callbacks
- **GLU** for helper functions such as gluPerspective and gluLookAt.

The project runs on native Linux. It does not support WSL or Virtual Machines.

Installation & Setup Instructions on Linux

Open terminal and run the following commands to install all required libraries:

```
sudo apt update
sudo apt install build-essential freeglut3-dev libglu1-mesa-dev mesa-utils unzip
```

Navigate to the directory containing the ZIP file:

```
unzip Group_7_Space_Shuttle_3D.zip
cd Group_7_Space_Shuttle_3D
```

From the project root directory where the **Makefile** is located, run:

```
make
```

This will compile the project using the provided Makefile and generate the executable. To run it:

```
./shuttle
```

To remove all compiled object files and the executable, run:

```
make clean
```

Note: This project is intended for **Linux only** and is not guaranteed to work on Windows or macOS

Using the Project:

The project supports both mouse and keyboard controls. The camera can be moved freely in the exterior view, and the model can also be viewed from the interior.

Mouse Controls:

- **Left mouse drag**: rotate the camera around the shuttle
- **Right mouse drag**: zoom camera
- **Middle mouse drag**: pan the view
- **Mouse scroll wheel**: zoom in and out

Keyboard Controls:

- **1 to 6**: switch between preset camera views
- **!, @, #, $, %, ^:** switch between ortho preset views
- **W**: toggle wireframe and solid view
- **A**: show or hide the axis lines
- **S**: toggle flat and smooth shading
- **L**: turn lighting on or off (interior and exterior)
- **G**: toggle background fog
- **F**: switch between exterior and interior camera modes
- **W, A, S, D, Q, E**: move in interior mode
- **T**: toggle flames of SRBs and SSME nozzles
- **O**: open or close the payload bay doors
- **H**: cycle through part highlighting (when not in wireframe mode)
- **R**: resets the camera view to start view (interior and exterior)
- **Esc**: exit to exterior / close

3. Functionalities Implemented

The following functionalities have been implemented in the space shuttle model.

Exterior 3D Model

The project models the full Space Shuttle launch stack, consisting of the Orbiter, the External Tank (ET), and two Solid Rocket Boosters (SRBs). All geometry was built using OpenGL primitives and no 3D model files were loaded.

Orbiter body: The fuselage is a tapered cylinder (wider at the aft, narrower toward the front) drawn using gluCylinder, capped at the front by a black nose cone. The crew cabin is a box shape protruding from the top of the forward fuselage, fitted with a grid of blue windshield panes. Both delta wings are drawn as swept trapezoidal flat polygons with slight thickness on each side of the fuselage. The vertical tail fin is also a flat polygon with thickness, placed at the aft end. Two OMS (Orbital Maneuvering System) pods sit on either side of the aft fuselage and are drawn as cylinders with dome caps. The three SSME (Space Shuttle Main Engine) nozzles are placed in a triangular cluster at the very back, each shaped as a flared frustum (bell shape) with nested annular disk rings at the exit plane to simulate depth.

External Tank and SRBs: The ET is an orange cylinder with a hemispherical dome cap on top, attached below and behind the Orbiter. The two SRBs are taller, narrower cylinders, each with an aft skirt section (a slightly wider cylinder at the base) and a cone nose cap. Structural attachment struts are drawn as thin boxes: two struts connect the ET to the Orbiter underside, and four struts per SRB connect each booster to the ET.

Fuselage text decal: The word "CS 302N" (configurable via config.h) is rendered on one side of the fuselage using a texture generated at runtime from a TrueType font. The texture is projected onto the port side using OpenGL texture matrix transforms, with blending enabled so the text appears as dark painted lettering over the white hull. (The text rendering was done using a header file called stb_truetype.h, can be found here.)

Materials and Lighting

Lighting setup. Three OpenGL light sources are configured. GL_LIGHT0 simulates the sun which is a bright white-yellow directional light. GL_LIGHT1 simulates the glow reflected off the Earth which is a dim blue-tinted fill light coming from below. GL_LIGHT2 is a warm point light placed inside the crew cabin, only active when the camera is in interior mode. It uses quadratic attenuation so its brightness falls off naturally with distance.

Material preses. Every surface has its own material definition specifying ambient, diffuse, specular, and shininess values. The white orbiter tiles have low specular (they look matte), while the SSME nozzles have high specular (dark steel with visible highlights). The ET is matte orange, the SRBs are light grey metallic, the windshields are glossy blue, and interior surfaces each have carefully tuned properties, for example, the instrument panel backing is a dark glossy plastic while the seat fabric is nearly specular-free. A small procedural foam texture was generated once and mapped onto the cylinder, dome, and cap of ET using GLU or manual coordinates. Texture matrix scaling (4×S, 2×T) controls tiling, and GL_MODULATE blends it with the base color without affecting other objects.

Shading and lighting toggles. The user can switch between smooth shading (normals interpolated across each polygon face, GL_SMOOTH) and flat shading (one normal per polygon face, GL_FLAT) using the S key. The lighting can be turned off entirely with the L key, which shows the model in its base colours without any light calculation.

Camera System

Exterior orbit camera: By default the camera orbits around the shuttle stack. Left mouse drag rotates the view (changes azimuth and elevation). Right mouse drag zooms in and out. Middle mouse drag pans the target point. The mouse scroll wheel also zooms. Elevation is clamped so the camera cannot flip past the top or bottom poles.

Camera presets with smooth interpolation: Pressing number keys 1 through 6 triggers a cinematic move to a preset viewpoint (front, left side, back, top-down, three-quarter, and a close-up aft view). Instead of snapping instantly, the camera smoothly glides to the target using exponential interpolation driven by a repeating timer. Any mouse input during the move cancels it and returns to normal orbit mode.

Interior first-person camera. Pressing F switches to a first-person camera placed inside the crew cabin. In this mode, WASD moves the camera forward, backward, left, and right along the cabin floor. Mouse movement looks around (yaw and pitch). The cursor is hidden and locked to the screen centre using glutWarpPointer, so the mouse never hits the window edge. Pitch is clamped to prevent the view from flipping upside down. Pressing Escape exits back to the exterior orbit camera. The field of view and near clip plane change when entering interior mode so the interior geometry is not clipped away.

Interior - Flight Deck and Mid-Deck

Flight deck: The flight deck is a fully enclosed room built from six wall panels (floor, left wall, right wall, aft wall, forward wall with the payload bay doors acting as the ceiling). Thin trim strips run along all wall-floor and wall-ceiling joints to hide seams. Inside the room there are two pilot seats, each with a seat pan, backrest, headrest, four metal legs, and armrests. In front of each seat is a T-shaped control yoke. The forward wall has a large instrument panel: a 4×6 grid of coloured rectangular gauges (each gauge has a bezel frame) backed by a dark panel board. Above the pilots, an overhead switch panel holds a grid of small coloured indicator lights with emissive glow. Side consoles sit against each wall with small button rows on top. A centre console with two throttle levers sits between the seats. The windshield is modelled from the inside as a framed array of glass panes divided by horizontal and vertical mullions.

Mid-deck: The mid-deck sits below the flight deck and is also a fully enclosed room. The left and right walls are covered in grids of storage lockers, each with a handle, label strip, and door seam lines. The floor has a grid of panel lines drawn as thin raised boxes. On the aft wall there is a ladder connecting up to the flight deck (two vertical rails and five rungs) and an equipment rack with shelf dividers. In the centre of the floor is a hatch cover with a frame surround and a handle.

Payload bay doors: Two large cargo doors are hinged along the top edges of the fuselage. Each door is split into an aft segment and a forward segment to clear the overhead switch panel. Pressing O starts a timer-driven animation that rotates the doors open (up to a configured maximum angle) or closes them again. The door state is shown in the HUD. Each door panel has transverse structural ribs and a longitudinal rib on its inner face.

Starfield and Background

A background of 2000 stars is generated once at startup and stored in a display list for fast rendering. Each star is placed at a random position on the surface of a large sphere using spherical coordinates with a fixed random seed (so the pattern is the same every time). Stars have slightly varied brightness. A small fraction are tinted warm (orange-red) or cool (blue-white) to add realism. It is visible through the open payload bay doors when in interior mode. The background also contains randomly placed meteors, asteroids and galaxies of varying shapes and sizes with a togglable fog. A moving animation was added to the meteors and asteroids as well.

HUD Overlay

At the end of every frame, a 2D overlay is drawn in screen space. The projection matrix is temporarily switched to orthographic (matching the window dimensions in pixels) so text can be placed at exact pixel positions. The top-left corner shows the current view mode (EXTERIOR, INTERIOR, or CINEMATIC), the lighting and shading state, the door state, and the active part highlight. The bottom-left corner shows a context-sensitive control legend where different keys are listed depending on whether the user is in exterior or interior mode. After drawing, the original projection matrix is restored and depth testing is re-enabled. All text is rendered using glutBitmapCharacter.

Part Highlighting

Pressing H cycles through the major parts of the shuttle: Orbiter, External Tank, Left SRB, Right SRB, SSME nozzle cluster, and OMS pods (then back to none). The selected part is drawn a second time as a yellow wireframe overlay on top of the normal geometry. This is done by binding a 1×1 yellow texture and drawing the part's geometry again in line polygon mode with polygon offset applied, so the yellow outline sits cleanly on top without Z-fighting. The current highlight name is shown in both the window title bar and the HUD.

Display Lists for Performance

The ET and both SRBs are compiled into OpenGL display lists once when the scene is first loaded. On every subsequent frame, drawing these components is just a single glCallList call instead of re-executing all the geometry commands. This significantly reduces CPU work per frame for the parts of the stack that never change shape. The starfield is also stored in a display list for the same reason.

Rendering Utilities

Primitives library: A shared set of geometry functions is used throughout: DrawBox (axis-aligned box with correct outward normals on all six faces), DrawCylinder (GLU frustum with optional end caps), DrawTexturedCylinder (same but with UV generation enabled for texture mapping), DrawDisk (flat annular ring), DrawQuad (arbitrary four-point polygon with automatic normal calculation), DrawDome (hemisphere built from triangle strips), and DrawText2D (bitmap string at a 2D screen position).

Scene reset: Pressing R resets both the scene state (doors closed, wireframe off, axes on, lighting on, smooth shading, no highlight) and the camera (back to the default exterior orbit position).

4. Output

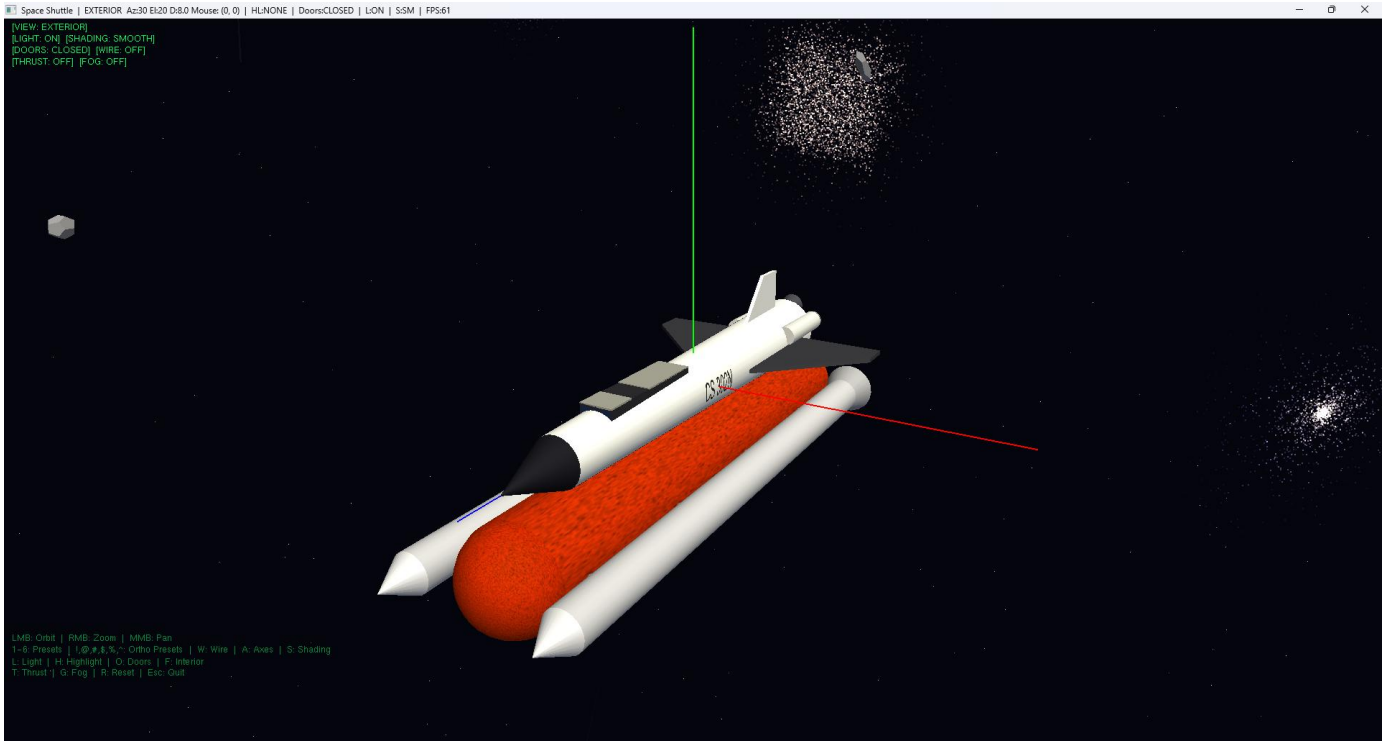

Figure 1: Default view

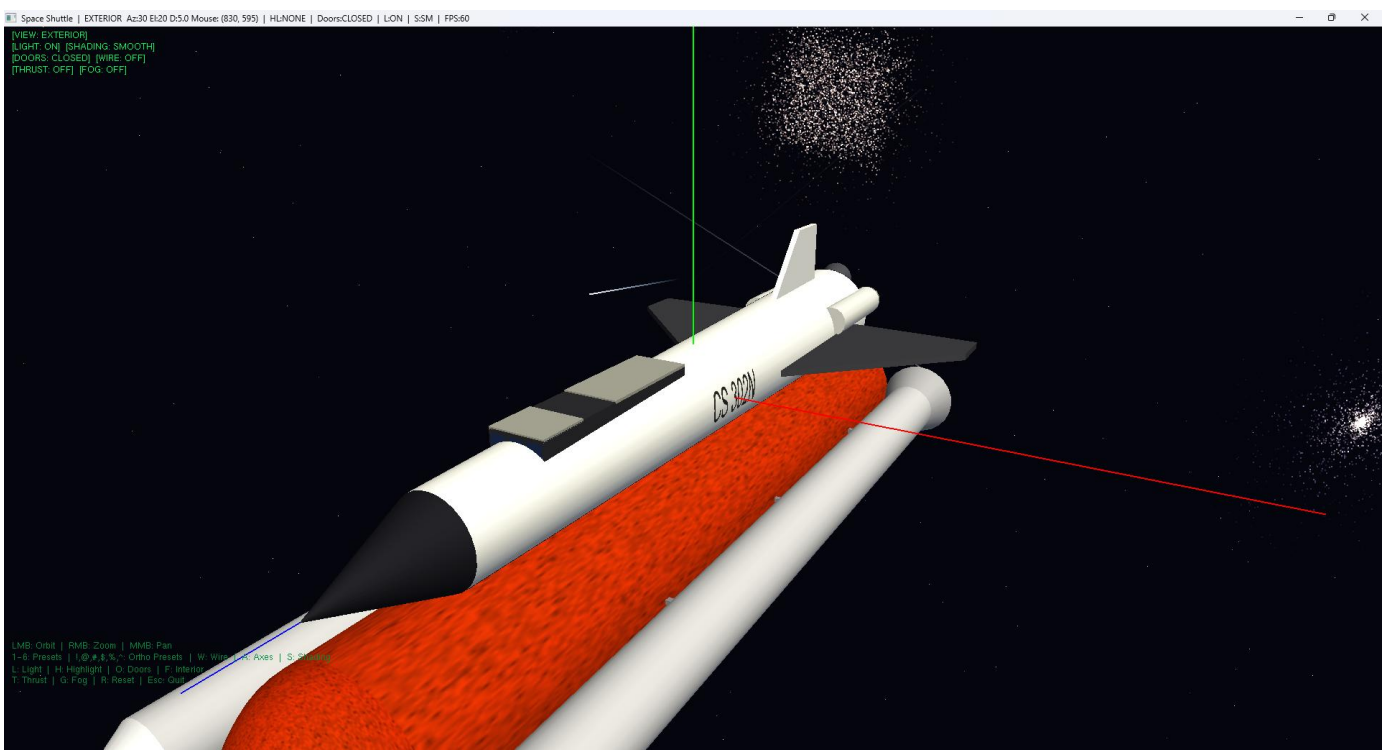

Figure 2: Zoomed In view

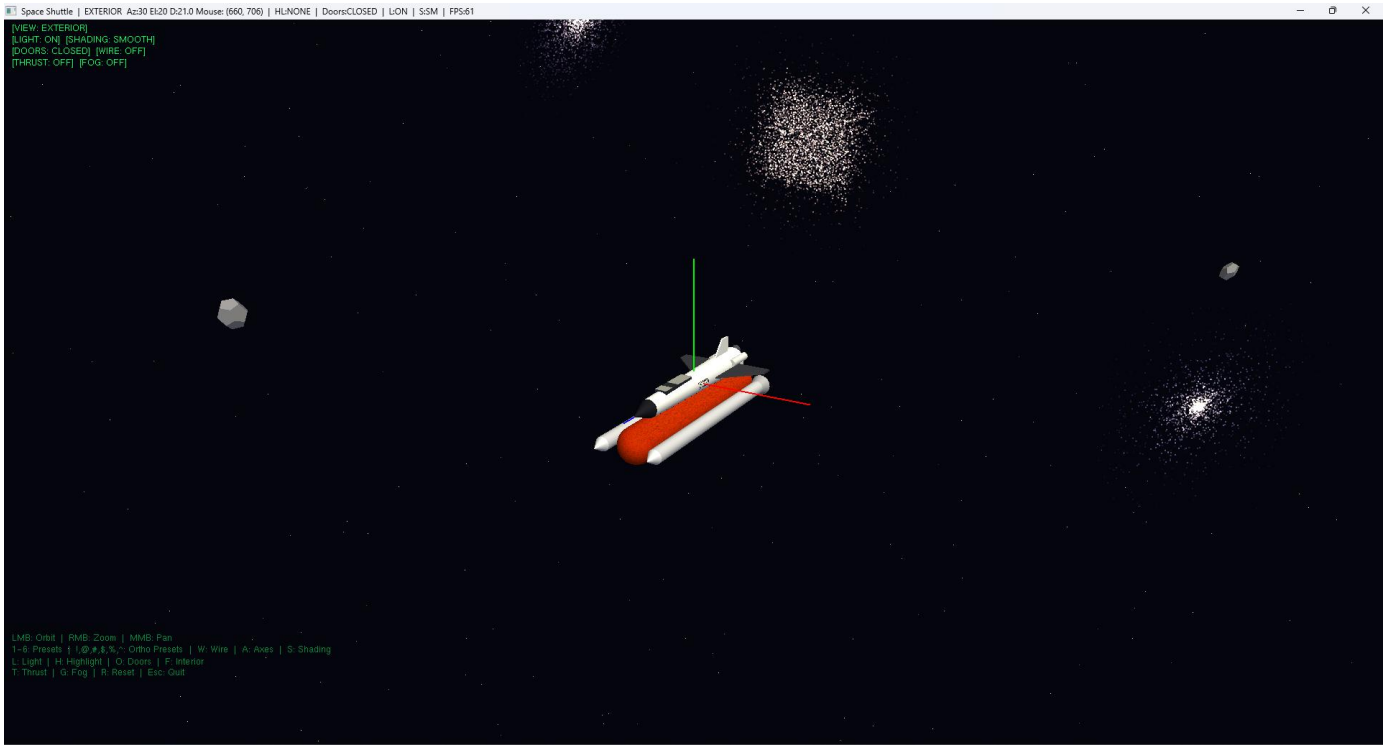

Figure 3: Zoomed Out view

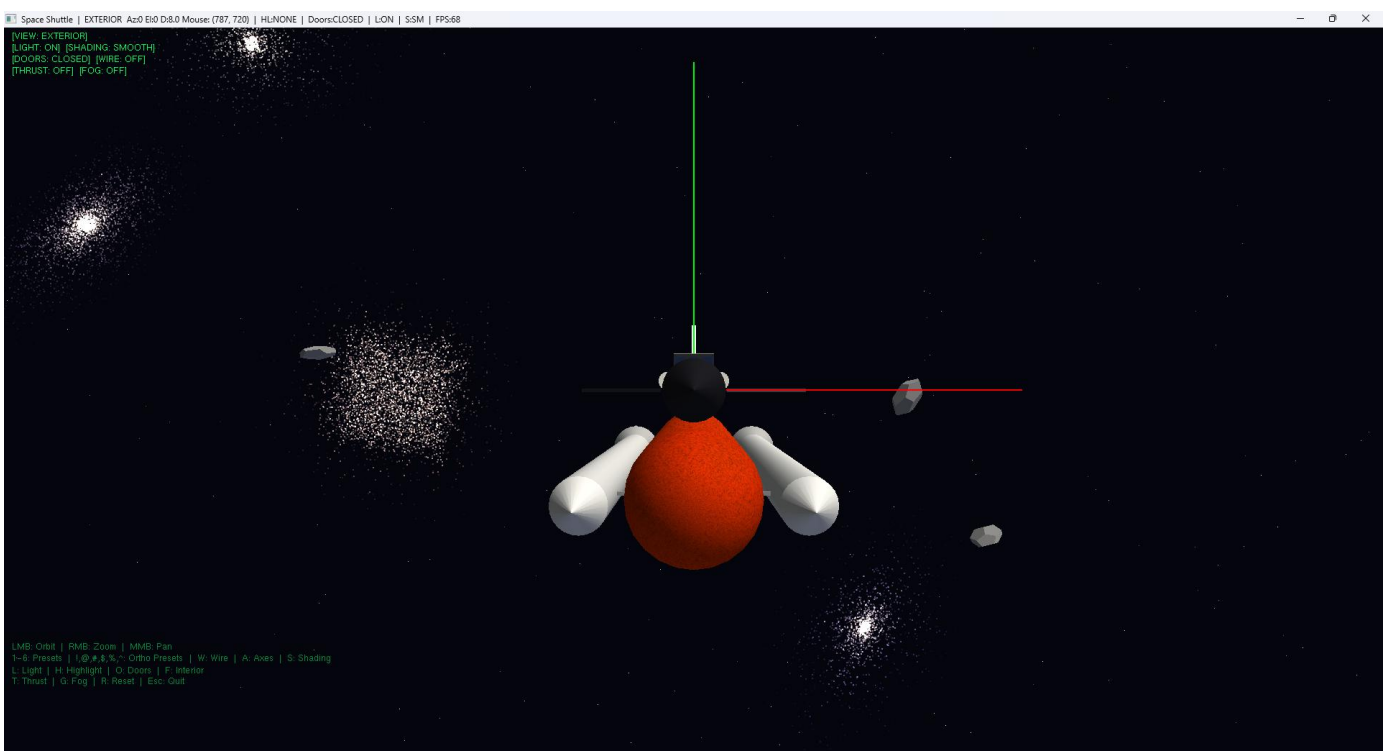

Figure 4: Preset 1 (Perspective Projection)

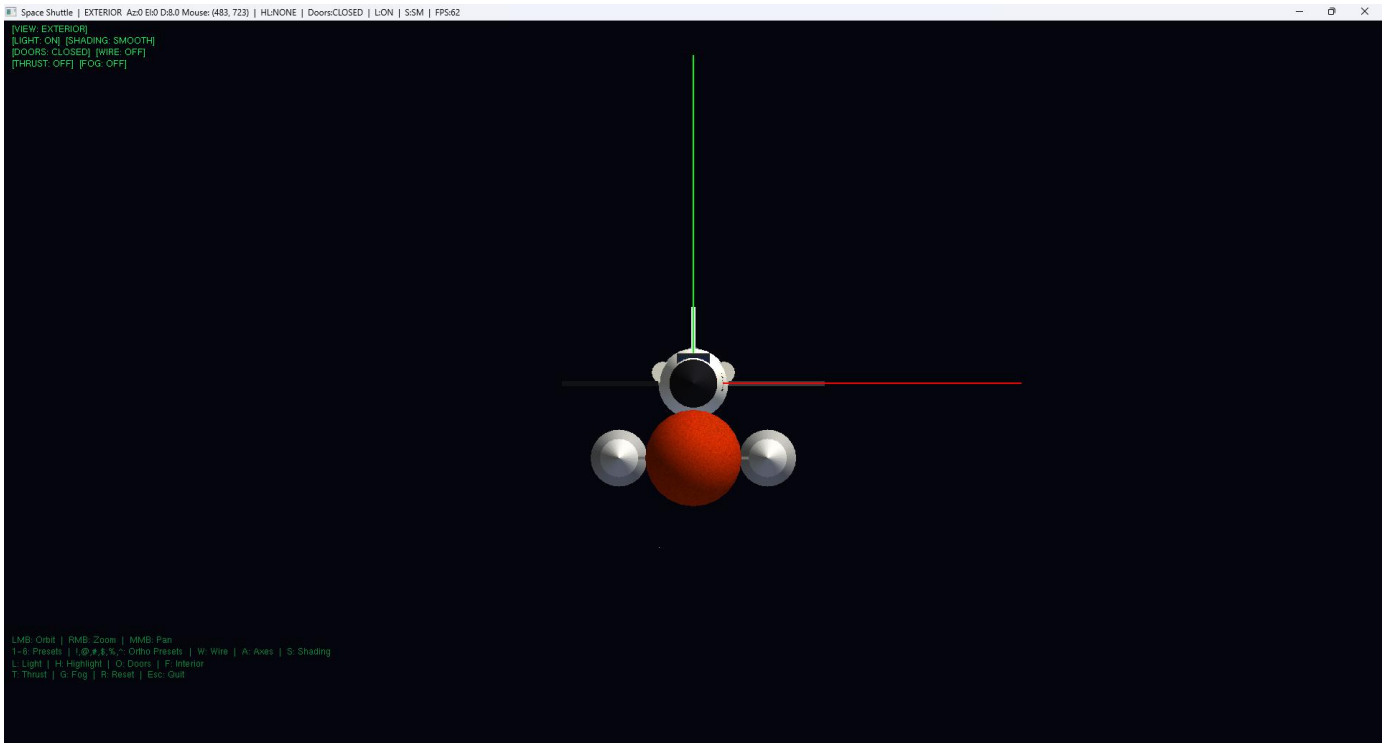

Figure 5: Preset 1 (Orthographic Projection)

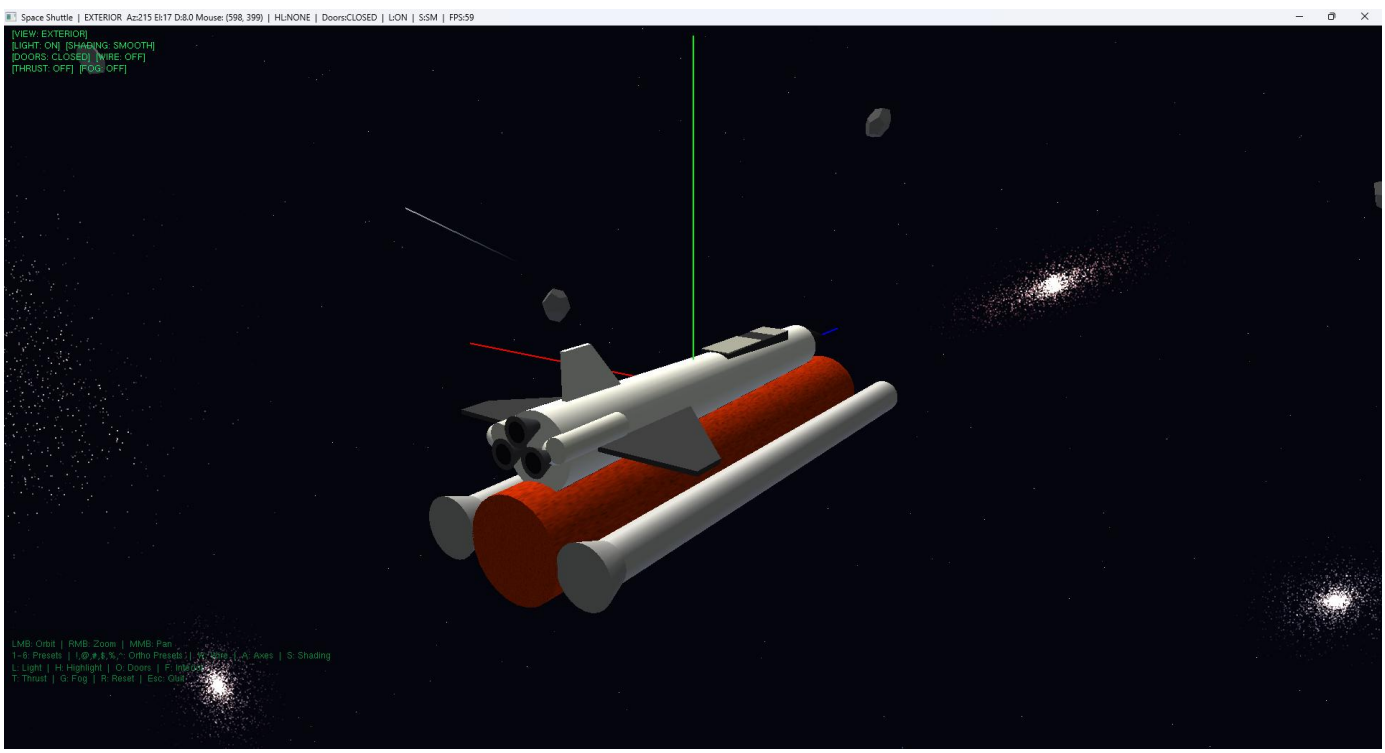

Figure 6: Back view

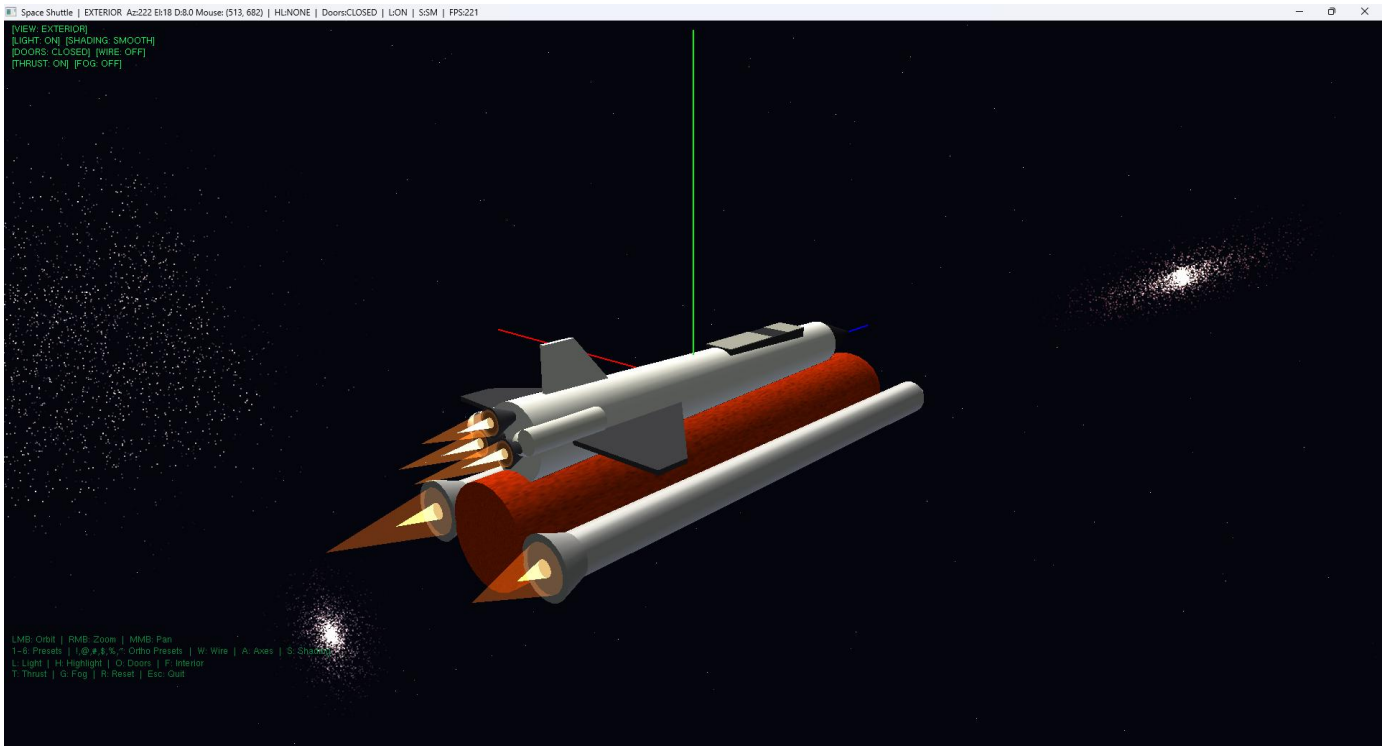

Figure 7: Thrusters toggled on

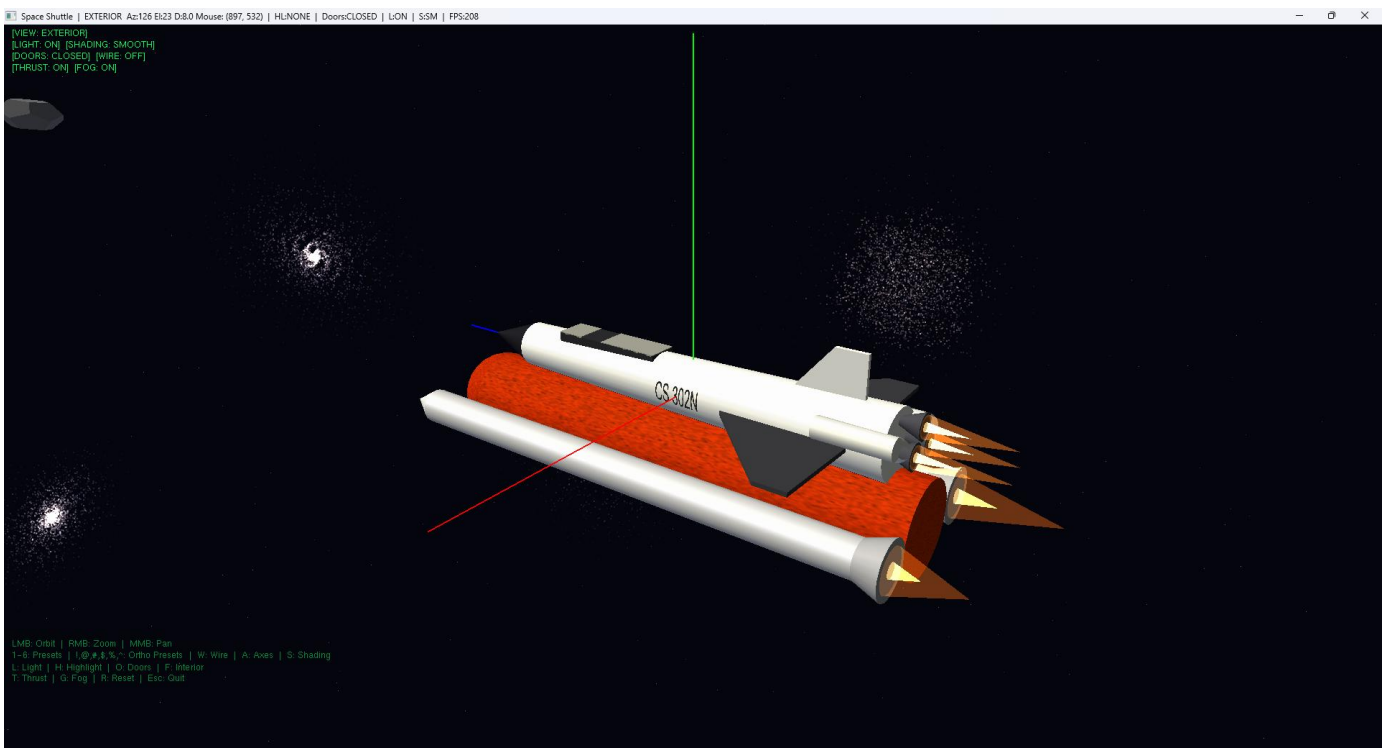

Figure 8: Fog toggled on

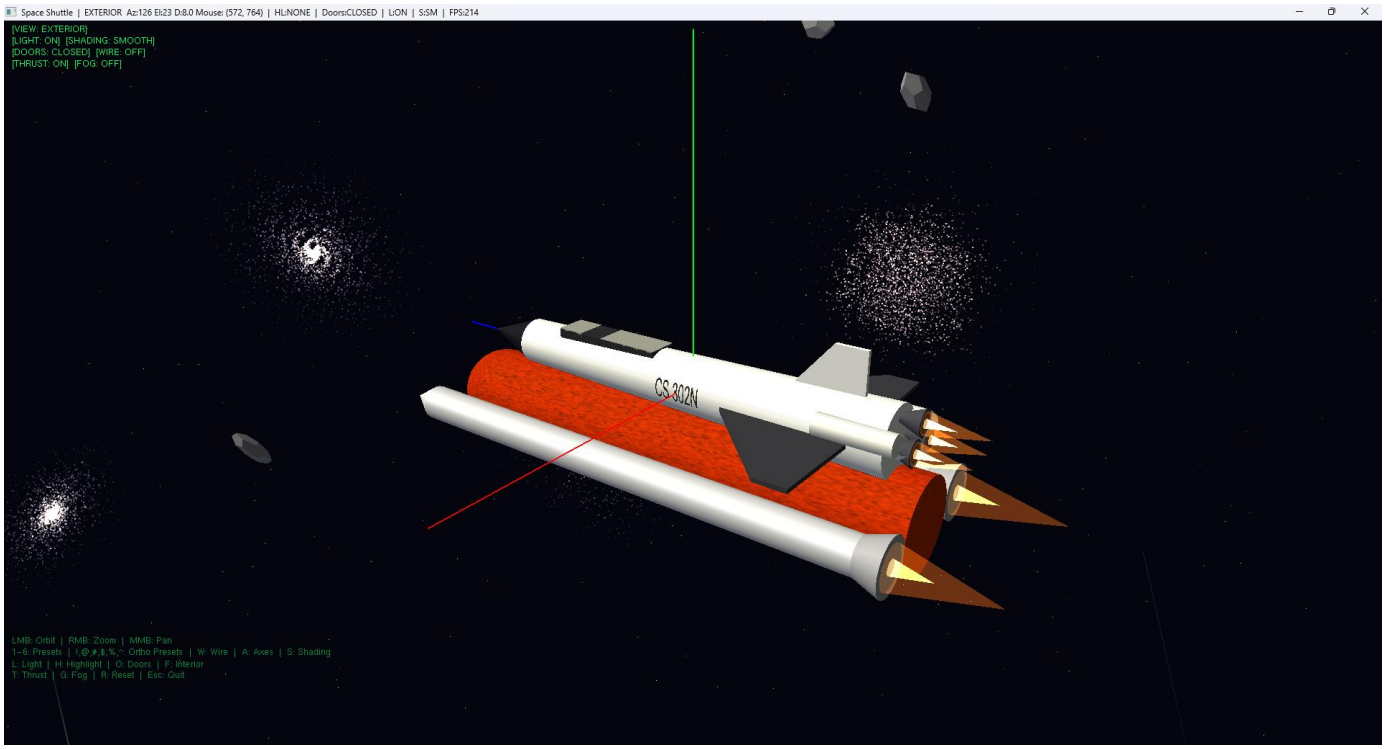

Figure 9: Fog toggled off

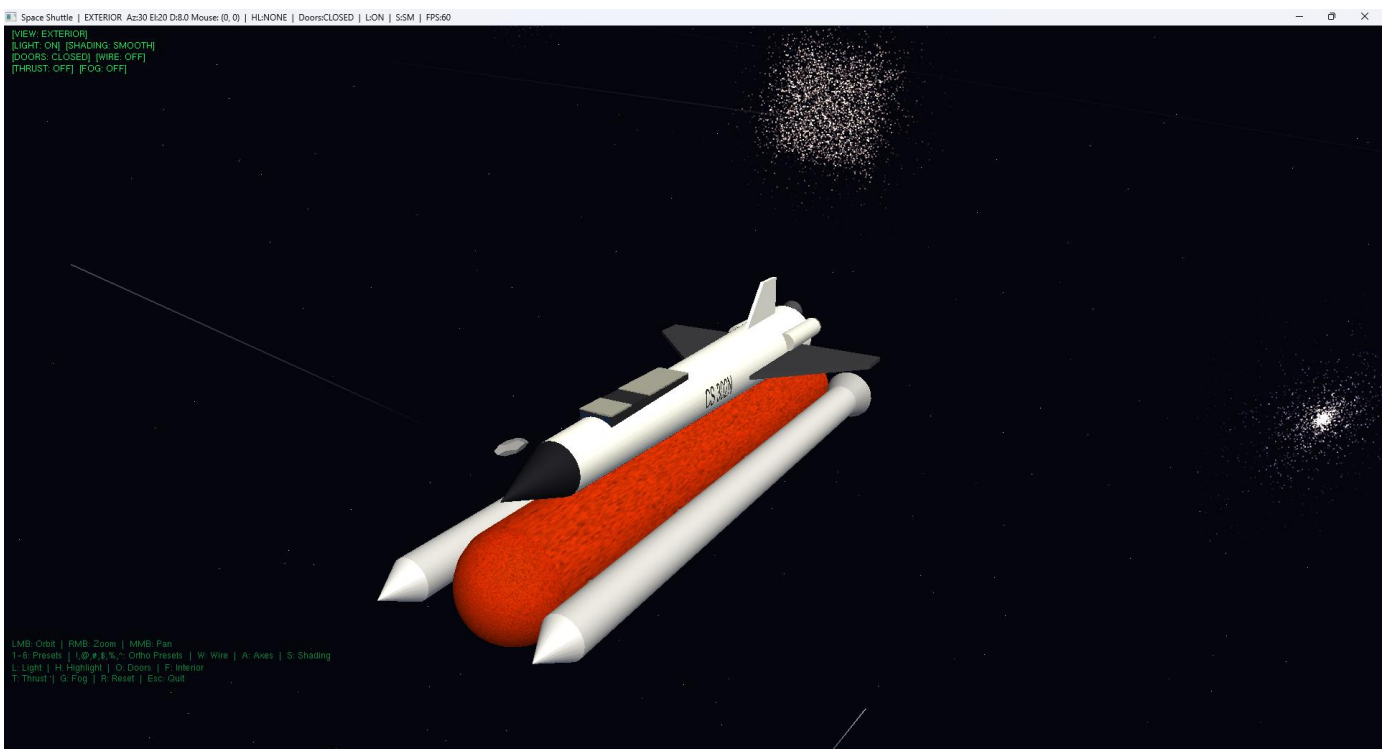

Figure 10: Axes toggled off

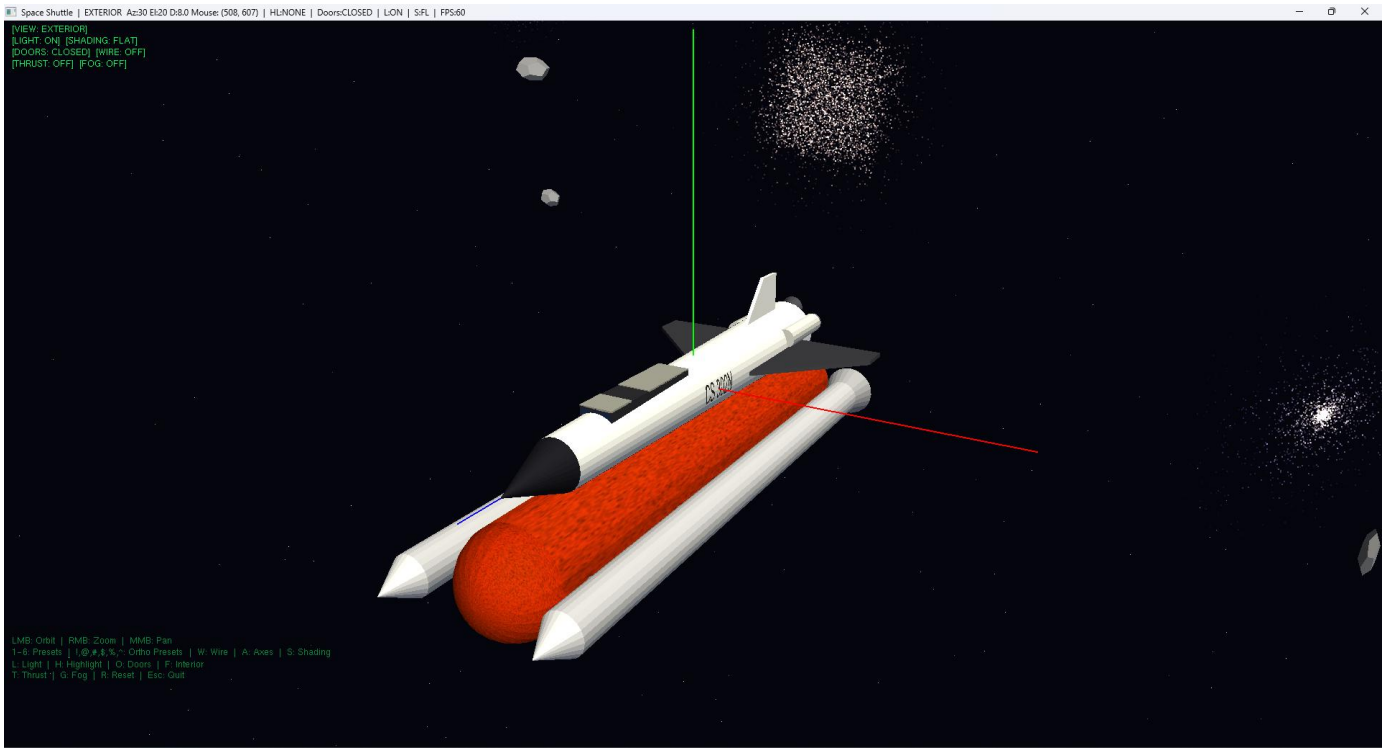

Figure 11: Flat Shading

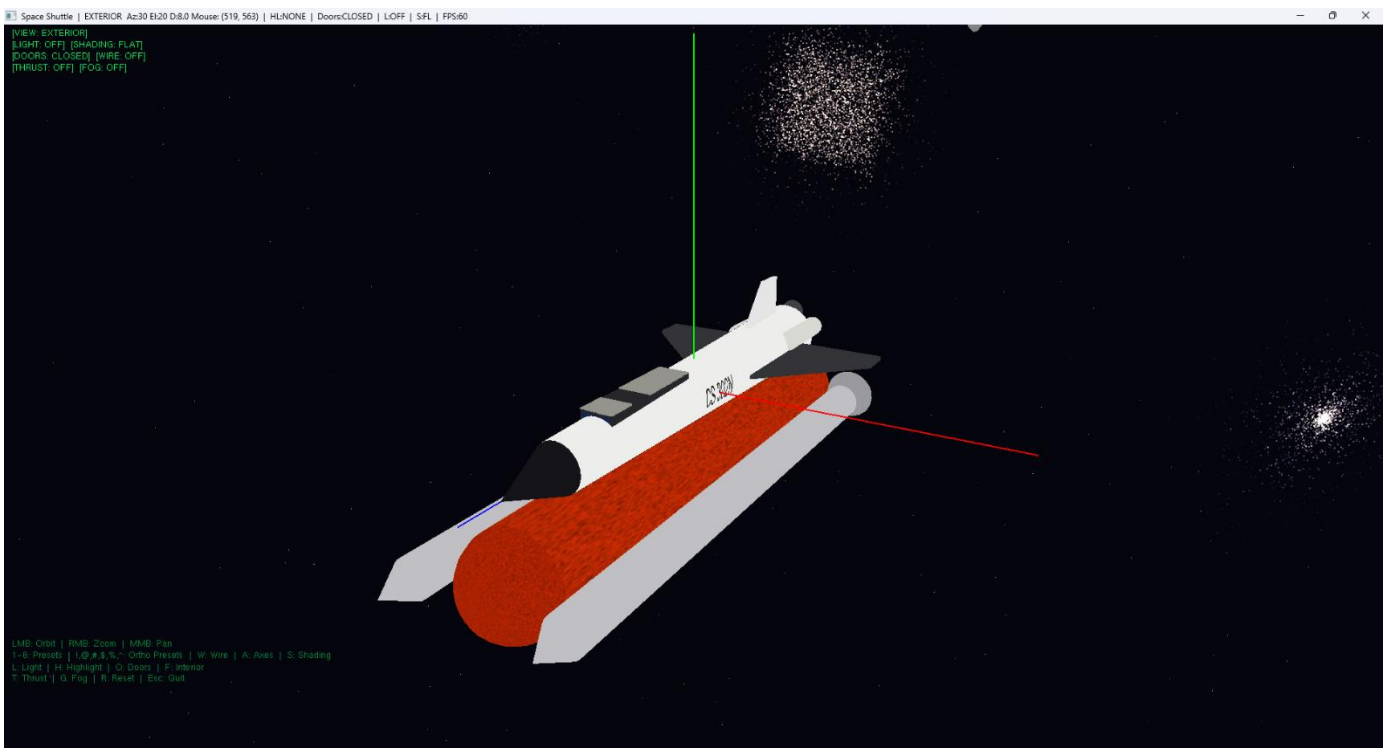

Figure 12: No lighting

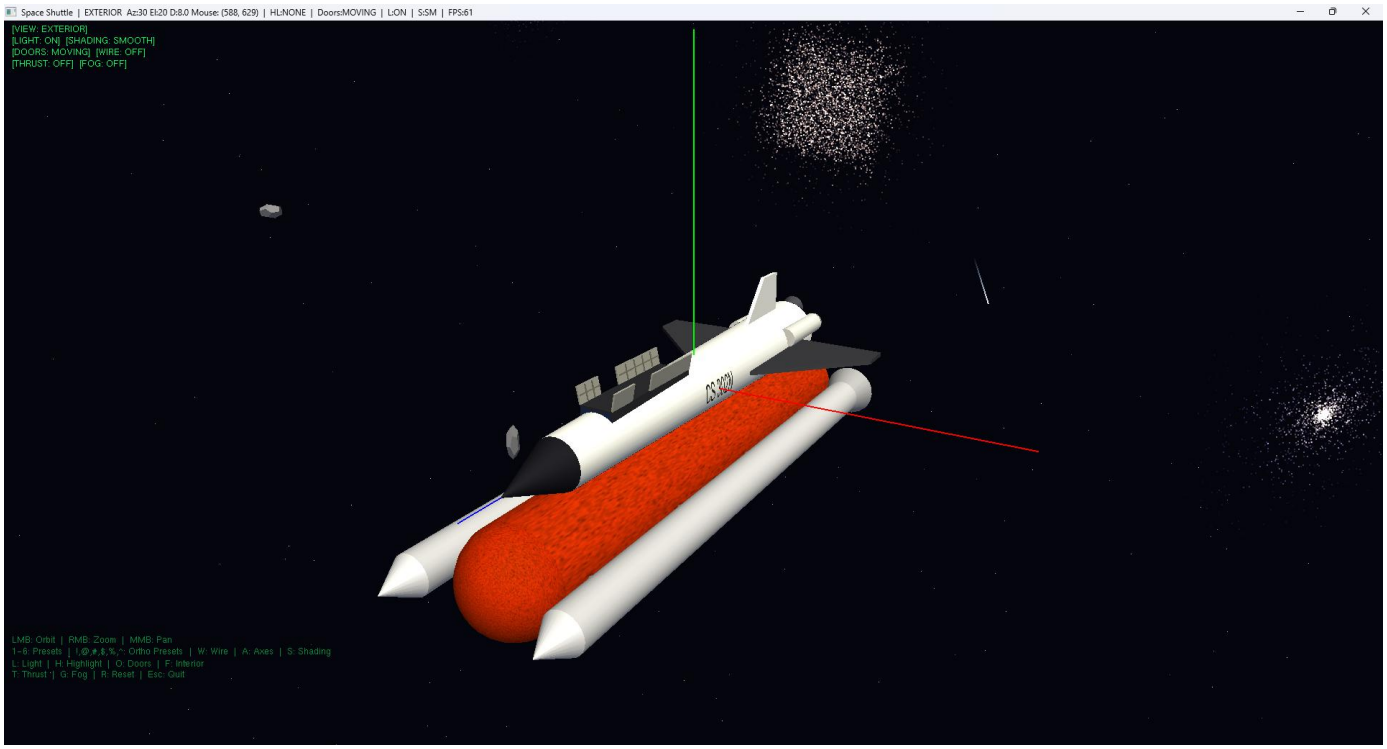

Figure 13: Door opening (mid-animation)

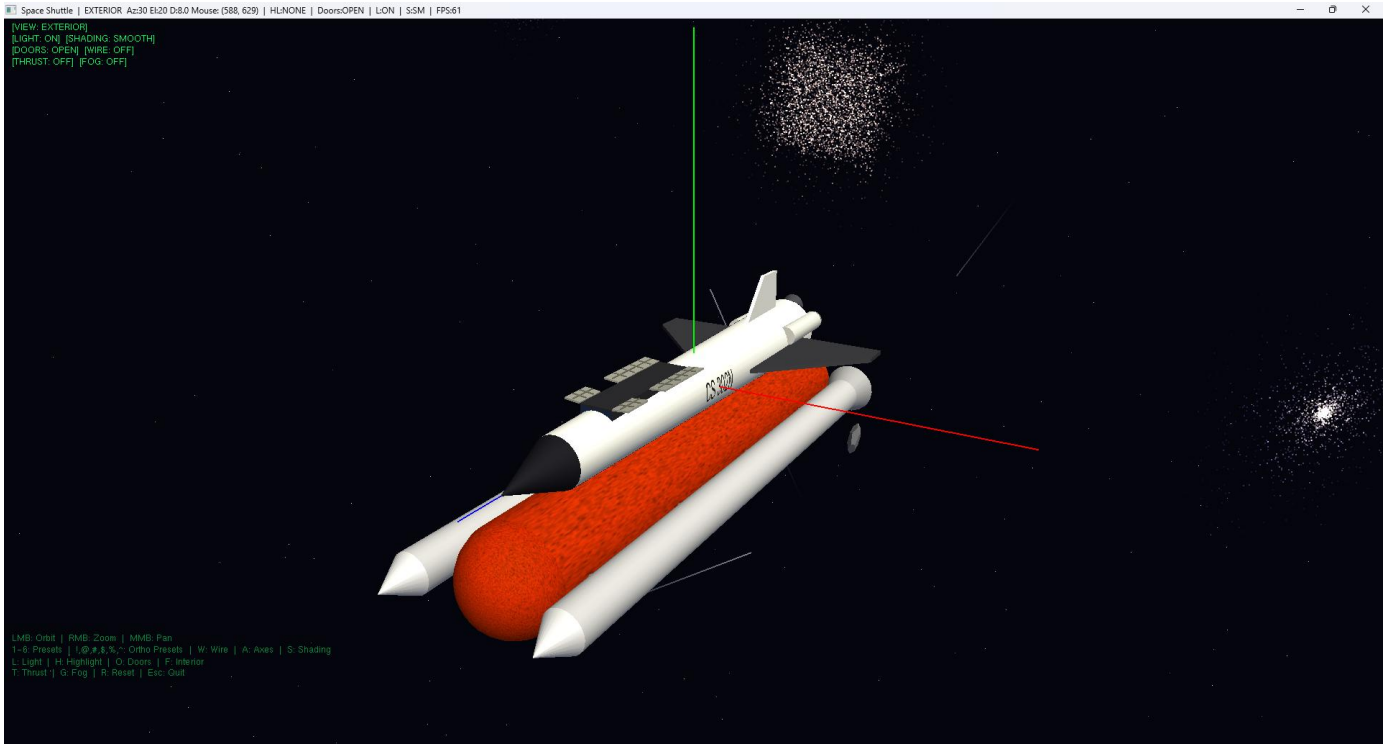

Figure 14: Door fully opened

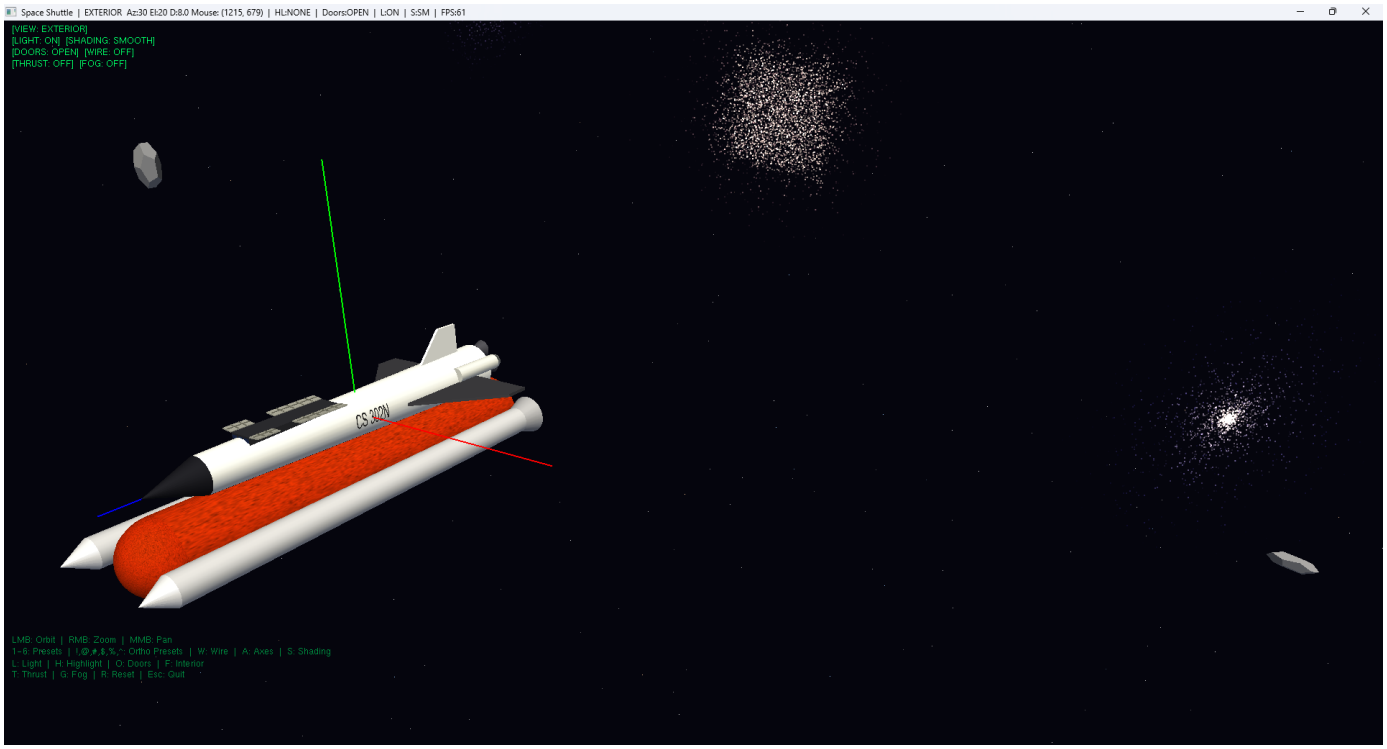

Figure 15: Panned view

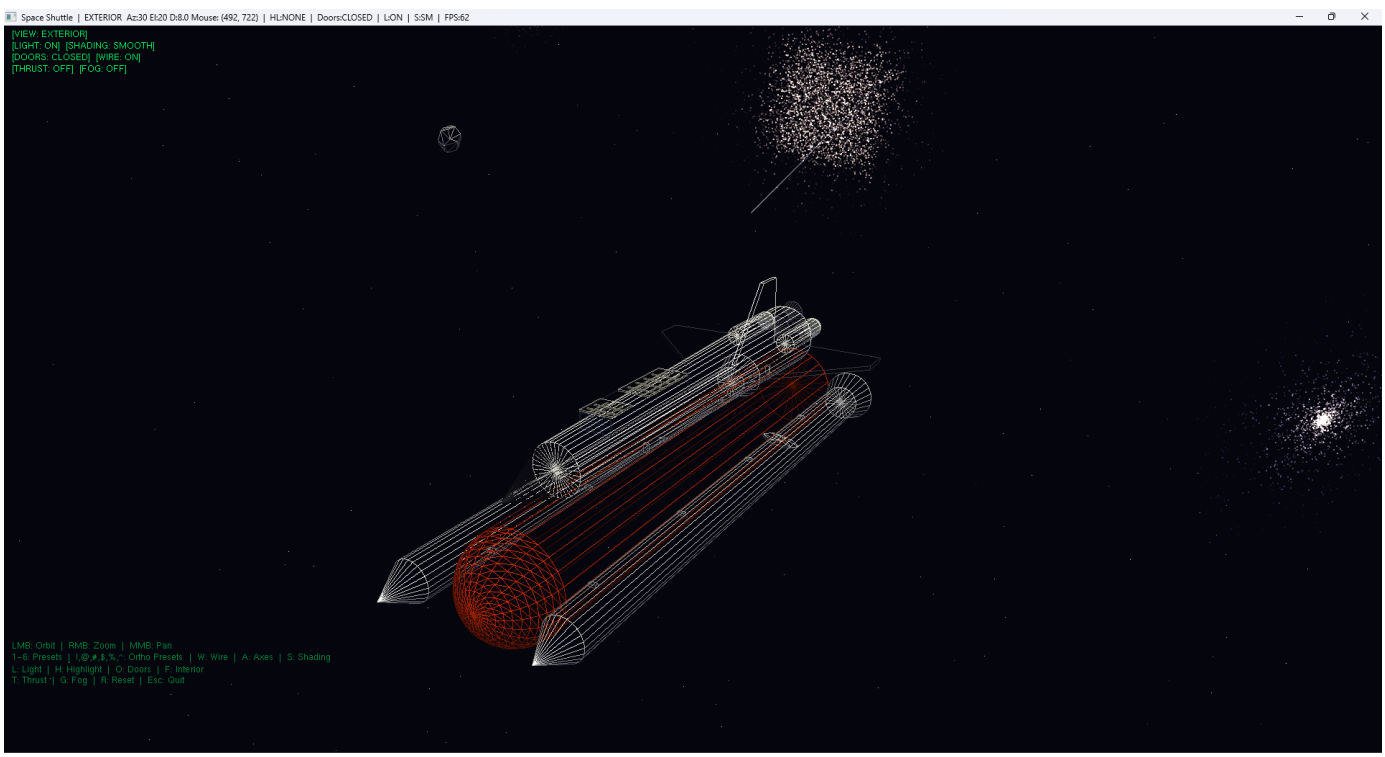

Figure 16: Wireframe mode toggled on

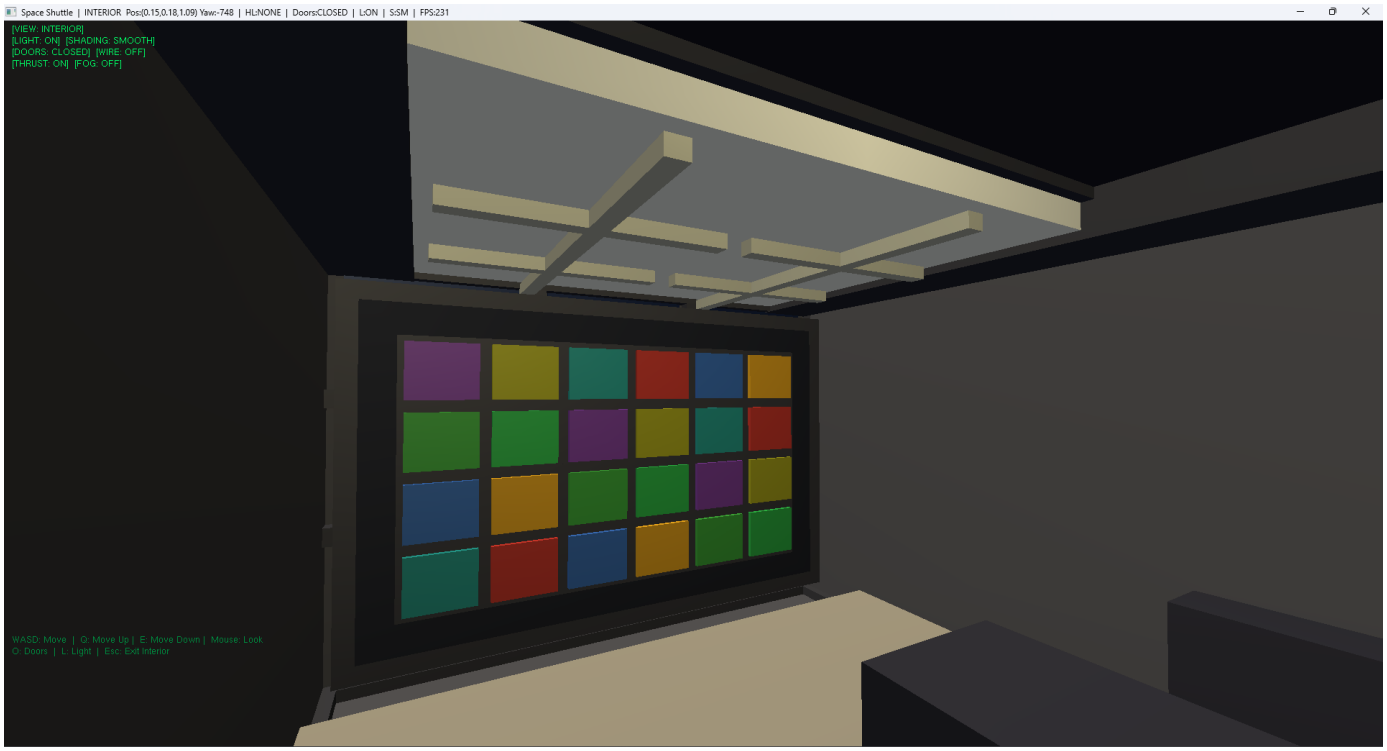

Figure 17: Cabin interior view (Doors closed)

Figure 18: Cabin seats view (Door closed)

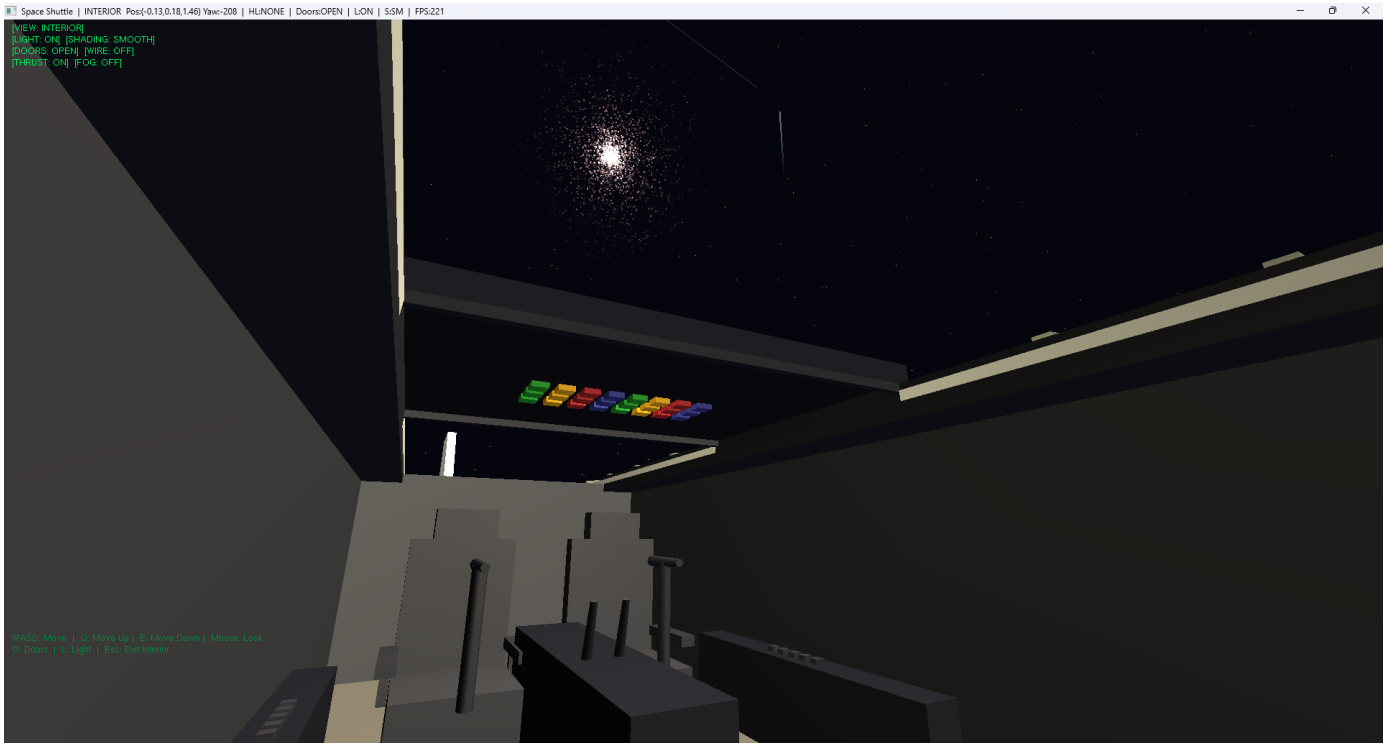

Figure 19: Cabin seats (Doors opened - Stars, galaxies and meteors visible)

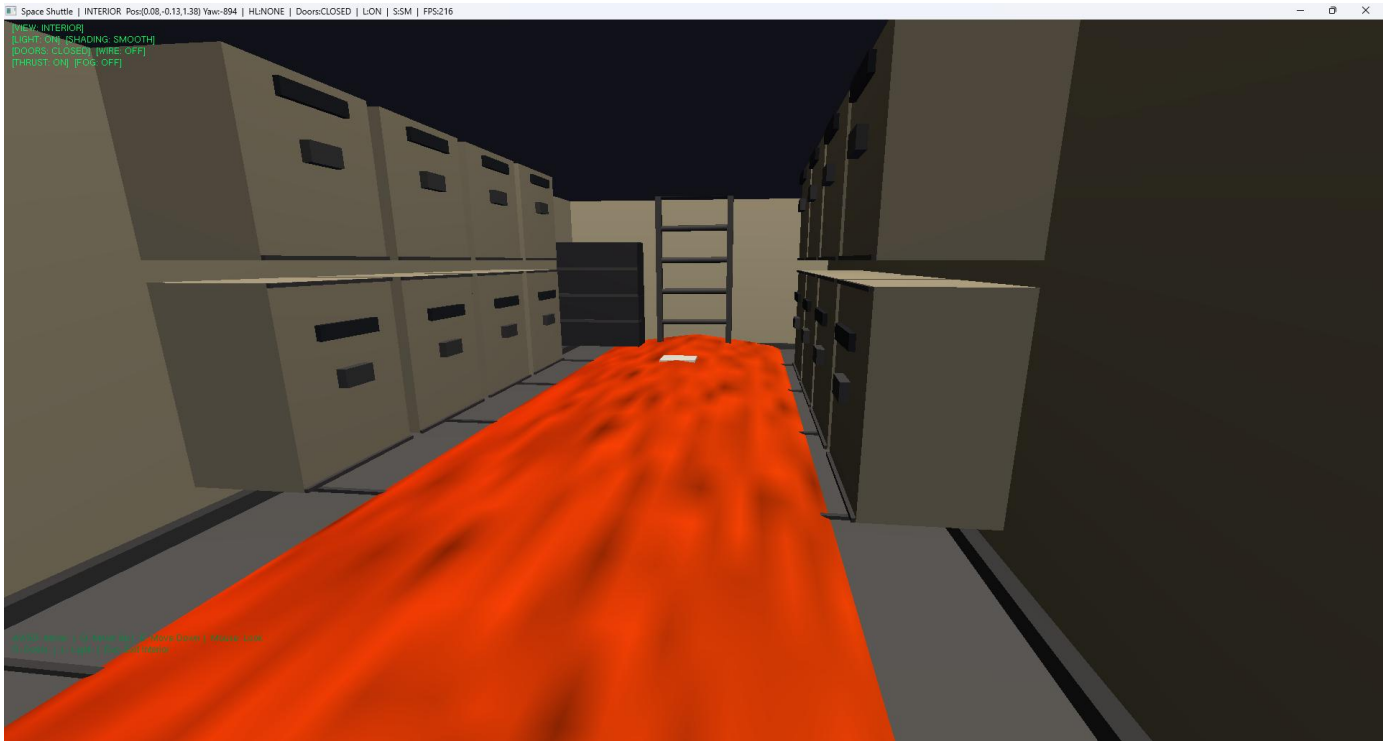

Figure 20: Mid deck locker room view